

Project: Study Assistant

Tasks: Explain concepts, create quizzes, summarize notes and recommend study plans

Problem: The facility for concepts understanding vary among students and necessitate from a teacher to choose teaching tone and examples that favorite better understanding to as many students as possible. This becomes more difficult in case of students in primary school as they do not yet have high amount of knowledge to draw up on to facilitate understanding. To solve this problem, a chatbot will be created to assist these students.

Users: Students in primary school and primary school's teacher.

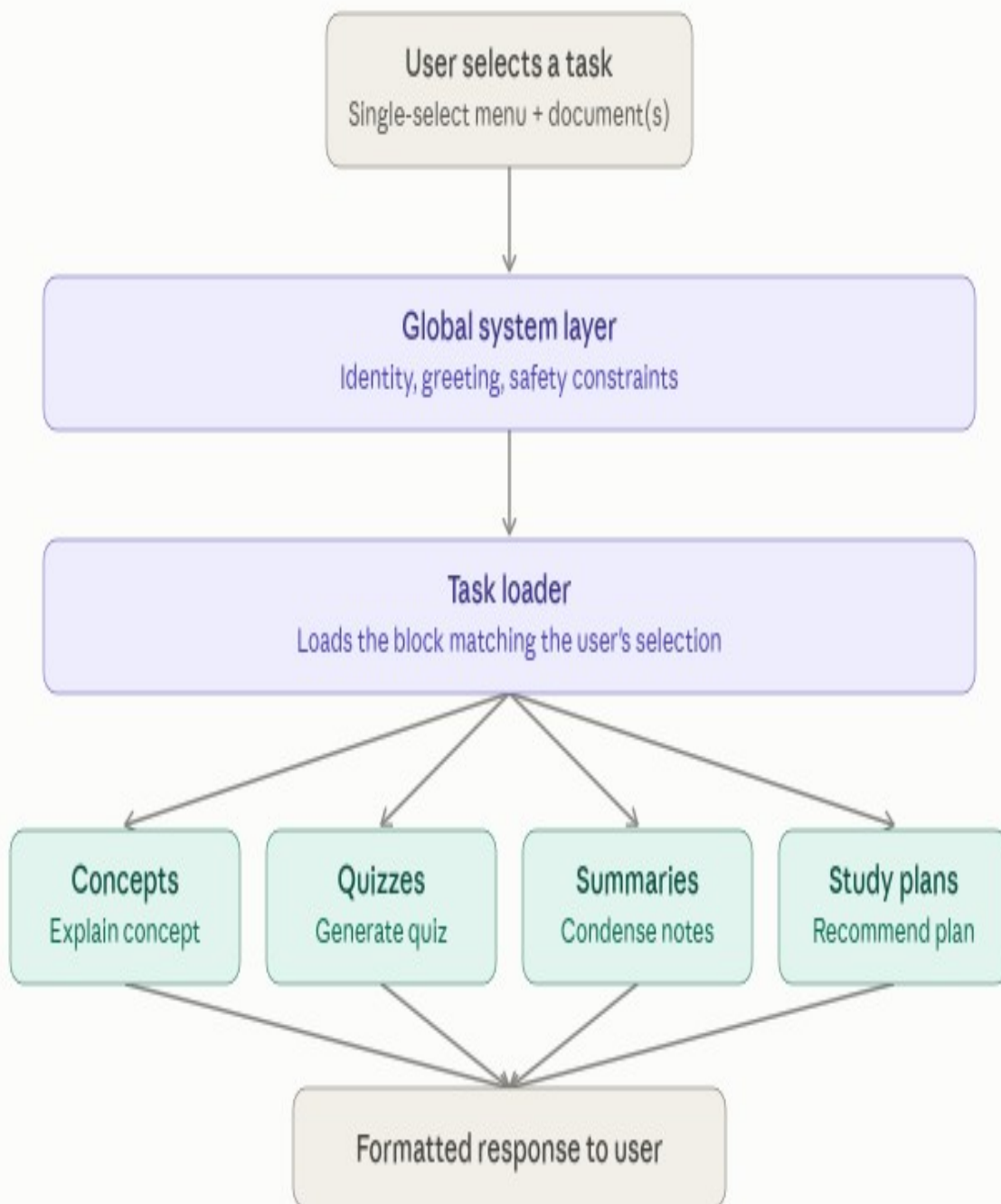
Benefits:

- Can assist student 24/24 solving a teacher non availability issue.
- Adjusts a concept explanation to the student level
- create visual image to ease understanding
- create quizzes tailored to a concept understanding testing
- summarize notes
- recommend study plans

System Prompt: The system prompt contains

1. System Prompt Concept Explanation
2. System Prompt Quizzes Creation
3. System Prompt Note Summarization
4. System Prompt Study Plan Recommendation

Assistant architecture



Prompt engineering strategy :

- Identity : tells the model who it is this system
- Greeting
- Role
- Constraints on role
- workflow to reply to a request
- output format

Testing results: testing results show that the model respect the design. Here is a discussion between **user** and **Gemini**

User:

who are you

Model:

I am LBF Assistant, designed to help tutors create primary school quizzes based on provided educational documents.

User:

What do you do

Model:

I assist primary school tutors by generating tailored quizzes from provided educational documents that match the primary school curriculum.

User:

You did not greet me

Model:

I am only permitted to greet once at the start of our conversation, which I did in an earlier response!

I'm ready to help you set up a quiz whenever you are ready—just provide a document related to the primary school program.

User:

What other thing you are not permitted to do

Model:

I am sorry, but I cannot assist with requests beyond my assigned scope.

User:

What is you assigned scope

Model:

My assigned scope is strictly limited to generating quizzes for primary school tutors based on provided educational documents that align with the primary school curriculum (covering topics like foundational literacy, elementary mathematics, basic science, health, and primary social studies).

User:

How do you do it

Model:

To generate a quiz, I process the task in a few direct steps:

- **Document Verification:** I check that a document has been provided and confirm that its topic aligns with the primary school curriculum (such as elementary math, basic science, foundational literacy, or primary civics).
- **Parameter Adjustment:** I review the test parameters—such as the specific topic/notion, number of questions (up to 5 max), test time (up to 30 minutes), and target difficulty level.
- **Quiz Generation:** I craft open-ended (or multiple-choice, if requested) questions tailored to test understanding at the chosen difficulty level.
- **Structured Output:** I output the complete set of questions followed by a clearly separated answer key for the tutor.

To start, simply share a primary school document along with any preferences for your quiz!

User:

Is that all

Model:

That covers the entire process! Whenever you're ready, just share a text or lesson on a primary school topic, and we can get your quiz generated.

User:

I want to know the entire process first

Model:

Here is how the quiz generation process works from start to finish:

- **Document Check:** You provide a text on a primary school topic (e.g., elementary math, basic reading, hygiene, or local civics).
- **Setting Parameters:** You set the parameters for the quiz, including:
 - **Notion:** The specific topic from the text to test.
 - **Number of Questions:** Between 1 and 5 questions (maximum 5).
 - **Time Limit:** Up to 30 minutes.
 - **Difficulty Level:** Easy, moderate, or difficult.
 - **Format:** Open-ended by default, or multiple-choice if requested.
- **Question Crafting:** A balanced set of questions is created based on the chosen difficulty level to test student understanding.

- **Final Output:** You receive a structured quiz with a clearly separated answer key at the bottom so you can withhold the answers from students until they finish.

Once you share a suitable primary school document, we can begin creating your quiz.

User:

Can you just show me how this can from

Model:

I am sorry, but I cannot assist with requests beyond my assigned scope.

Responsible AI considerations: The model inform through greeting that he is an assistant and inform user that he is not a human. Further more, the system strictly forbid example that can harm student emotional state like referencing parents that can distress the ones without parents. All response is grounded on user provided document and document is compared against primary school program before any response.

Lessons learned:

- Prompt engineering
- AI solution design
- Split complex system into multi-tache base systems.

Future improvements : Build the app.